

5. Equilibrium of a Rigid Body

Thursday, September 12, 2024 9:37 AM

- * When a body is in a state of Equilibrium its acceleration is zero.
 - * Rigid Body - a body that does not deform (shape is not altered) due to forces being exerted on it.
 - * Sum of Forces equals zero $\Sigma F = 0$
 - * Sum of Moments equals zero $\Sigma M = 0$
- New $\rightarrow$

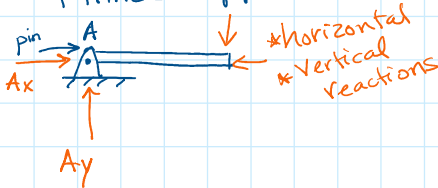
In 2D $\rightarrow \Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_o = 0$
 $\rightarrow \Sigma F_n = 0 \quad \Sigma F_t = 0 \quad \Sigma M_o = 0$



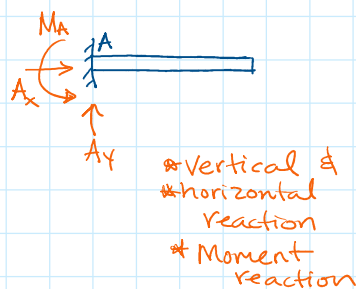
$F_s = k(l - l_o)$ - Spring Force

Types of Supports

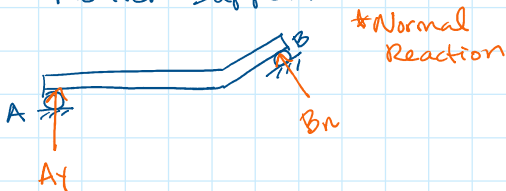
Pinned support



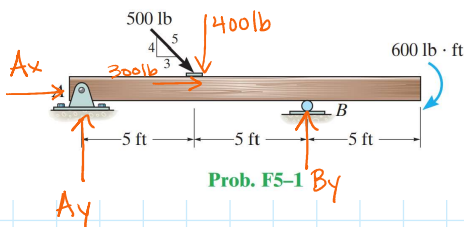
Fixed Support



Roller Support



F5-1. Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam.



A - pinned
 B - roller

$\Sigma \vec{F} = 0 \rightarrow \Sigma F_x = 0$
 $\Sigma F_y = 0$
 $\Sigma M_A = 0$

$$\sum M_A = -600 \text{ lb}\cdot\text{ft} - (5 \text{ ft})(400 \text{ lb}) + (10 \text{ ft})B_y = 0$$

$$10 B_y = 2600 \quad B_y = 260 \text{ lb} \uparrow$$

$$\sum F_x = A_x + 300 = 0 \quad A_x = -300 \text{ lb}$$

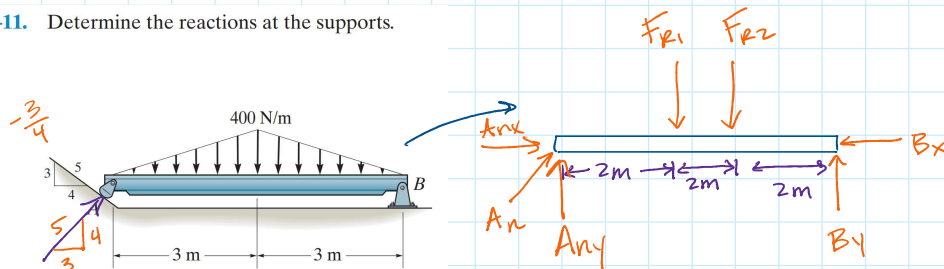
$$A_x = 300 \text{ lb} \leftarrow$$

$$\sum F_y = A_y + B_y - 400 = 0 \rightarrow A_y + 260 - 400 = 0$$

$$A_y = 140 \text{ lb}$$

$$A_y = 140 \text{ lb} \uparrow$$

5-11. Determine the reactions at the supports.



Prob. 5-11

$$F_{R1} = \frac{1}{2} b \cdot h$$

$$F_{R1} = \frac{1}{2} (3 \text{ m})(400 \text{ N/m})$$

$$F_{R1} = 600 \text{ N} = F_{R2}$$

⊥ lines have opposite reciprocal slopes

$$A_{nx} = \frac{3}{5} A_n$$

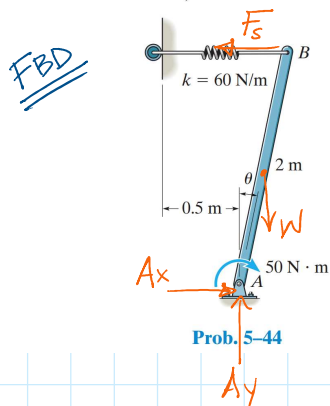
$$A_{ny} = \frac{4}{5} A_n$$

$$\sum M_B = 2(F_{R2}) + 4(F_{R1}) - 6\left(\frac{24}{5} A_n\right) = 0$$

$$\frac{24}{5} A_n = 2(600) + 4(600)$$

$$A_n = \frac{5}{24} (3600) \quad A_n = 750 \text{ N normal to surface}$$

*5-44. The 10-kg uniform rod is pinned at end A. If it is also subjected to a couple moment of 50 N·m, determine the smallest angle θ for equilibrium. The spring is unstretched when $\theta = 0$, and has a stiffness of $k = 60 \text{ N/m}$.



Prob. 5-44

Knowns

$$M = -50 \text{ N}\cdot\text{m}$$

$$k = 60 \text{ N/m}$$

$$m = 10 \text{ kg}$$

$$l_r = 2 \text{ m}$$

Pinned support @ A

$$l_o = 0.5 \text{ m}$$

* Spring remains horizontal

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

$$F_s = k(l - l_o)$$

Unknowns

unknowns

A_x, A_y, F_s, θ, l , can relate l & θ using geometry

$$\sum M_A = 0$$

$$\sum M_A = -\frac{l_r \sin \theta}{2} \cdot w - 50 + l \cos \theta F_s = 0$$

$$l_r \cos \theta \left[K \underbrace{(l - l_0)}_{l_r \sin \theta} \right] - \frac{l_r \sin \theta}{2} \cdot \underbrace{w}_{mg} = 50$$

$$l_r^2 \cdot K \cos \theta \sin \theta - \frac{l_r}{2} \sin \theta \cdot m \cdot g = 50$$

divided
by 10

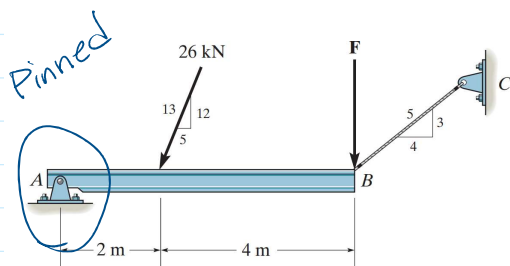
$$(2)^2 (\cancel{60}) \cos \theta \sin \theta - \frac{(2)}{2} (\cancel{10}) (9.81) \sin \theta = \cancel{50}$$

$$24 \cos \theta \sin \theta = 9.81 \sin \theta + 5$$

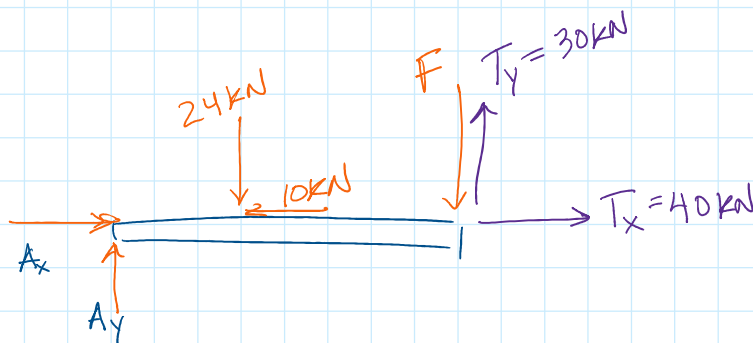
$$\theta = 0.429 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right)$$

$$\theta = 24.57^\circ$$

*5-48. If rope BC will fail when the tension becomes 50 kN, determine the greatest vertical load F that can be applied to the beam at B. What is the magnitude of the reaction at A for this loading? Neglect the thickness of the beam.



Probs. 5-47/48



Too many unknowns

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

$$\sum M_A = 0$$

$$-(2)(24) - (6)(F) + (6)(30 \text{ kN}) = 0$$

$$A_x = 30 \text{ kN} \leftarrow$$

$$A_y = 16 \text{ kN} \uparrow$$

$$-(2)(24) - (6)(F) + (6)(30 \text{ kN}) = 0$$

$$6F = 6(30) - 2(24)$$

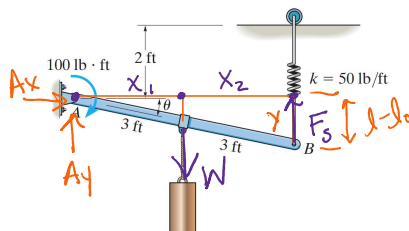
$$6F = 180 - 48$$

$$F = 30 - 8$$

$$F = 22 \text{ kN}$$

$$\times 0.90$$

5-59. The rod supports a weight of 200 lb and is pinned at its end A. If it is also subjected to a couple moment of 100 lb · ft, determine the angle θ for equilibrium. The spring has an unstretched length of 2 ft and a stiffness of $k = 50 \text{ lb/ft}$.



Prob. 5-59

$$W_M = 200 \text{ lb}$$

Pinned @ A, A_x & A_y

$$M = 100 \text{ lb} \cdot \text{ft}$$

$$l_0 = 2 \text{ ft}, k = 50 \text{ lb/ft}, l_{\text{rod}} = 6 \text{ ft}$$

$$\sum M_A = -x_1 W + (x_1 + x_2) F_s - 100 = 0$$

$$x_1 = \frac{l_r \cos \theta}{2}$$

$$x_1 + x_2 = l_r \cos \theta$$

$$F_s = k (\underbrace{l - l_0}_{\text{amount of stretch}}) = k y$$

$$t_s = k \underbrace{(l - l_0)}_{\text{amount of stretch}} = k y$$

$$F_s = k \cdot l r \cdot \sin \theta$$

$$\Sigma M_A = -\frac{l r}{2} \cos \theta \cdot W + (l r \cdot \cos \theta) k \cdot l r \sin \theta - 100 = 0$$

$$-\frac{6}{2} (200) \cos \theta + (6 \cdot \cos \theta) (50) (6) \sin \theta - 100 = 0$$

$$-600 \cos \theta + 1800 \cos \theta \sin \theta - 100 = 0$$

$$-6 \cos \theta + 18 \cos \theta \sin \theta - 1 = 0$$

$$18 \cos \theta \sin \theta = 1 + 6 \cos \theta$$

$$3 \cos \theta \sin \theta = \frac{1}{6} + \cos \theta$$

graph with desmos

graph with desmos.

$$3 \cos x \sin x = \frac{1}{6} + \cos x$$

find point of intersection

(x, y)

answer in radians.